GWO/PID Approach for Optimal Control of DC Motor

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***Abstract*— The work deals with application of Grey Wolf Optimization (GWO) for optimal speed control of DC motor. GWO is a bio inspired heuristic algorithm. Here, integral of time multiplied absolute error (ITAE) has been taken as an objective function for tuning the parameters of PID controller by GWO. Comparison of proposed GWO/PID scheme with other existing techniques has also been shown. It has been observed that proposed GWO/PID approach with ITAE as an objective function gives comparable overshoot and other parameters such as; settling and rise times are less when compared with existing approaches in the literature. The robustness and comparative analysis of proposed GWO/PID approach has also been carried out with variations in the parameters of DC motor.**

***Keywords*—DC Motor, PID-Controller, Grey Wolf Optimization, Optimal Control, ITAE.**

I. INTRODUCTION

Mainly two types of dc motors are used in the industries. The first one is conventional dc motor where flux is produced by the current through the field coil of stationary pole structure. The second type is brushless DC Motor (BLDC Motor) where the permanent magnet provides necessary air gap flux instead of the wire-wound field poles [1]. Due to the advantages of smaller volume, high force and simple structure of BLDC motor, it is widely applied in the areas which needs high performance drive [1].

The PID controllers maintain the output at a level so that there is no difference between the process variable and set point. PID controllers are broadly used in industrial plants due to their robustness and ease of implementation. Various algorithms are available in literature to tune the parameters of PID controllers, such as; Ziegler Nichols, Cohen-coon tuning and Z-N step response, etc. But, all of these classical methods have some limitations [2]. Also, Genetic Algorithm (GA) [3], Particle Swarm Optimization (PSO) [1, 4], Invasive Weed Optimization (IWO) [5], and SFS algorithms [6-7] are already available in the literature to tune the parameters of PID controller for speed control of DC motor.

The present work deals with application of GWO algorithm in tuning the parameters of PID controller for speed control of DC motor, where in ITAE has been used as an objective function. GWO is a bio inspired heuristic algorithm inspired by both the social hierarchy of wolves as well as their hunting behavior. The search starts with population of randomly generated wolves (solutions) in GWO. During hunting (optimization) process, these wolves estimate the prey's (optimum) location through an iterative procedure [8-11].

II. DC MOTOR-BASIC CONCEPTS

The DC motor converts DC energy into mechanical energy [12-15]. DC motor generates torque directly from the DC power supply to the motor by using internal communication, stationary permanent or electromagnets, and rotating electrical magnets. For simulation, the parameters and their values for DC motor used in present work have been given in Table 1 [5-7].

Table 1: DC motor parameters

Parameter	Value
Armature Ckt. Resistance (R_a)	0.4 ohm
Armature Ckt. Inductance (L_a)	2.7 H
Moment of Inertia (J)	0.0004 kg.m^2
Daming Ratio (D)	0.0022 N.m.sec/rad.
Armature Constant (K)	15 e-3 Kg.m/A
Motor Constant (K_b)	0.05 V.sec

Now, the objective is to obtain speed of DC motor close to the ideal/set point state. Figure 1 shows the equivalent circuit of DC motor with a PID controller.

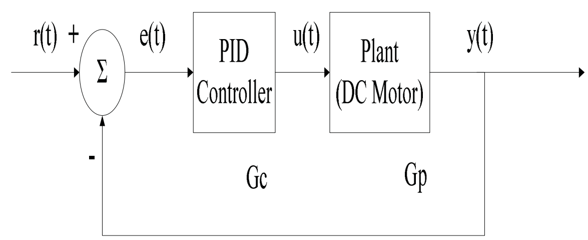

Fig. 1. Equivalent circuit of DC motor with PID controller

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III. PROBLEM FORMULATION

In general, the Eq. of PID controller is given as :

$$G_C = K_P + \frac{K_I}{s} + K_D s \tag{1}$$

For obtaining the unknown parameters of PID controller in (1) for speed control of DC motor to ideal/set point state, the fitness/objective function taken is ITAE and error is the output velocity of DC motor. Here, ITAE has been taken as an objective function because it gives smaller overshoots and oscillations than the other performance indices [7]. This ITAE is given by :

$$ITAE = \int_0^\infty t\left|e(t)\right|dt \tag{2}$$

The complete simulink model of DC motor with PID controller with ITAE as fitness function has been shown in Fig. 2.

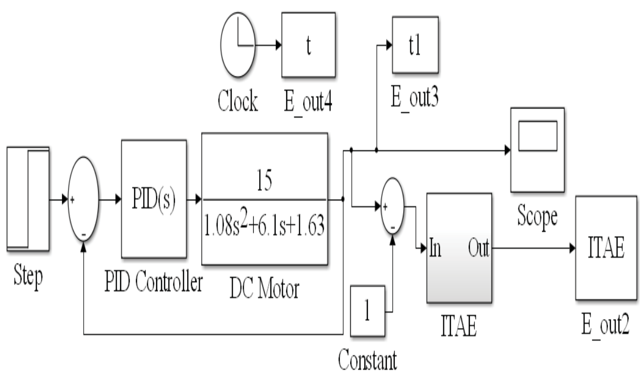

Fig. 2. Complete simulink model of DC motor with PID controller

IV. GREY WOLF OPTIMIZATION

Similar to the social hierarchy of grey wolves, there are four groups defined in GWO algorithm namely [8-10] :

- **Alpha (α) -** The leaders are a male and female, called alphas. The alpha is mostly responsible for making decisions about hunting, sleeping place, time to wake, and so on.
- **Beta (β) -** The second level in the hierarchy of grey wolves is beta. The betas are subordinate wolves that help the alpha in decision-making or other pack activities.
- **Omega (ω) -** The lowest ranking grey wolf is omega. The omega plays the role of scapegoat. Omega wolves always have to submit to all the other dominant wolves. If a wolf is not an alpha, beta, or omega, he/she is called subordinate (or delta in some references).
- **Delta (δ) -** Delta wolves have to submit to alphas and betas, but they dominate the omega. Scouts, sentinels, elders, hunters, and caretakers belong to this category.

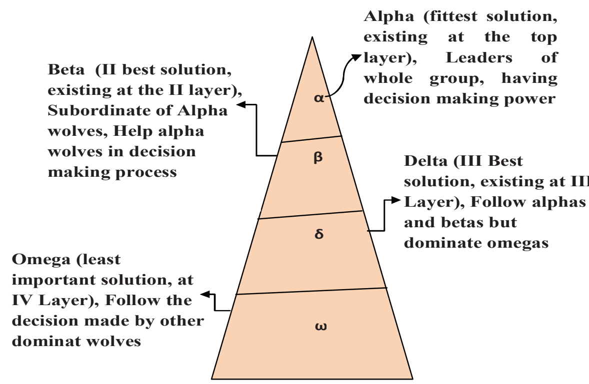

Fig. 3. Social hierarchy of GWO and functions of each group

The functions of each group have also been defined in Fig. 3 [10-11]. The pseudo code of GWO has been given in Appendix.

V. IMPLEMENTATION OF GWO/PID APPROACH

The GWO algorithm has been run in Matlab for the simulink model shown in Fig. 2 for 30 iterations and obtained parameters of PID controller are given by :

$$G_C = 6.8984 + \frac{0.5626}{s} + 0.9293\,s \tag{3}$$

For obtaining the above parameters of PID controller, the convergence of objective function; ITAE by GWO is shown in Fig. 4.

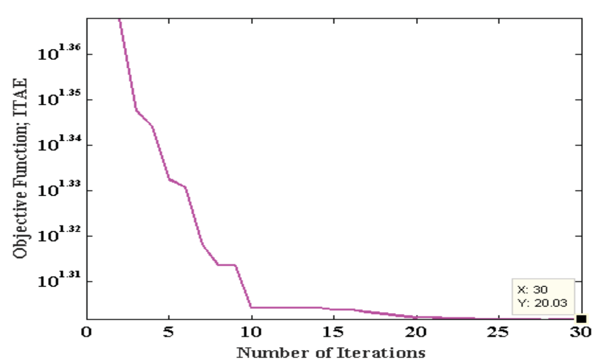

Fig. 4. Convergence of objective function

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In Table 2, the parameters of PID controller obtained by other existing techniques in literature [5, 7] for the same DC motor have also been given.

Table 2: Parameters of PID controller for DC motor obtained by GWO, IWO, PSO and SFS

Algorithm	K_p	K_I	K_D
GWO (Proposed)	6.8984	0.5626	0.9293
IWO [5]	1.5782	0.4372	0.0481
PSO [5]	1.5234	1.3801	0.0159
SFS [7]	1.6315	0.2798	0.2395

In Fig. 5, comparison of speed of DC motor without and with PID controller tuned by GWO with ITAE as an objective function is shown.

The comparison of speed of DC motor with other existing approaches has also been shown in Fig. 6. It can be seen in Fig. 6 that, GWO/PID approach with ITAE gives less settling and rise times in comparison to existing approaches. In Table 3, comparative analysis of proposed GWO/PID scheme with IWO [5], PSO [5] and SFS [7] has also been shown in terms of response's parameters. It can be seen in Table 3 that, the proposed GWO/PID approach gives less settling and rise times in comparison to existing techniques.

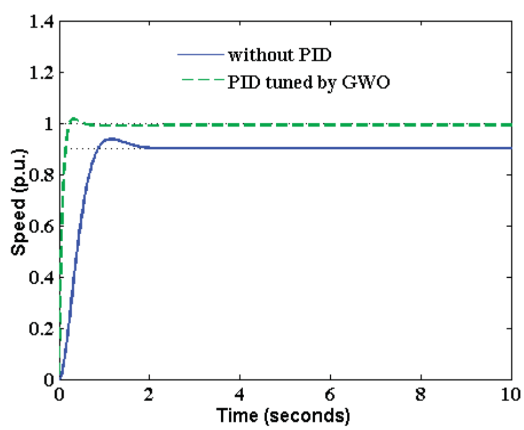

Fig. 5. Speed comparison of DC motor without and with PID controller

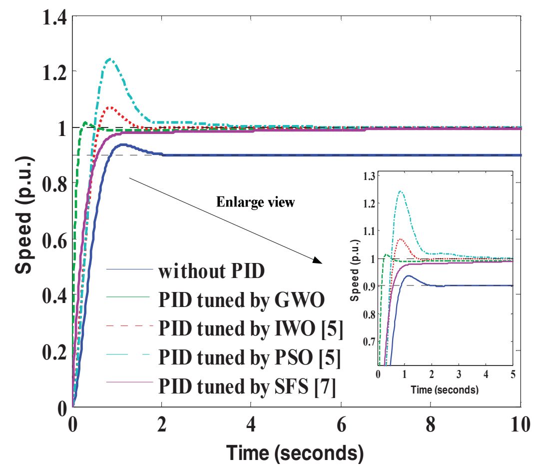

Fig. 6. Speed comparison of DC motor without and with PID controller with other approaches

Table 3: Comparison of response's parameters

Algorithm	Over Shoot (%)	Settling Time (sec)	Rise Time (sec)
GWO	1.51	0.205	0.139
IWO [5]	6.98	1.25	0.419
PSO [5]	24.2	1.8	0.356
SFS [7]	0	1.45	0.544

VI. ROBUSTNESS ANALYSIS

Table 4: Operating points

Case No.	R_a	K
1	0.4	0.015
2	0.2	0.012
3	0.1	0.014
4	0.3	0.015

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For evaluating the performance of the proposed GWO/PID scheme, different operating points in time domain have been tested. For this, operating points of Table 4 according to the changes in electrical resistance and K parameter in DC motor have been used [5, 7] and thereafter comparative analysis has been carried out. These operating points are shown in Table 4.

For all the above cases, the PID controller of (3) has been used which is obtained by GWO with minimization of objective function; ITAE. Now, the above operating points have been considered separately and simulation results have been shown.

The comparison of step responses has been shown in Fig. 7 and comparative analysis of proposed GWO/PID scheme with IWO [5], PSO [5] and SFS [7] has been shown in terms of response's parameters for operating point 1 in Table 5.

Figure 8 shows the comparison of step responses for operating point 2. In Table 6, comparative analysis of proposed GWO/PID scheme with IWO [5], PSO [5] and SFS [7] has been shown in terms of response's parameters for operating point 2.

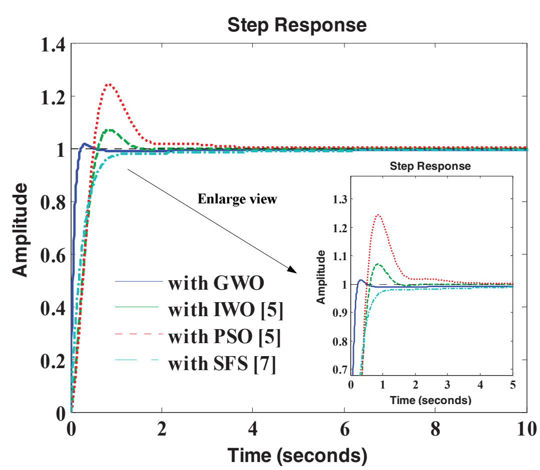

Fig. 7. Comparison of step responses for operating point 1 ; R_a = 0.4 ; K = 0.015

Table 5: Comparison of response's parameters for operating point 1; $R_a = 0.4$; $K = 0.015$

Algorithm	Over Shoot (%)	Settling Time (sec)	Rise Time (sec)
GWO (Proposed)	1.51	0.205	0.139
IWO [5]	6.98	1.25	0.419
PSO [5]	24.2	1.8	0.356
SFS [7]	0	1.45	0.544

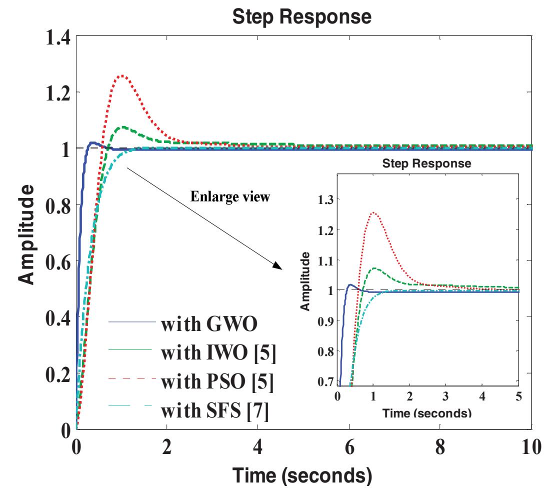

Fig. 8. Comparison of step responses for operating point 2; R_a = 0.2 ; K = 0.012

Table 6: Comparison of response's parameters for operating point 2; $R_a = 0.2$; $K = 0.012$

Algorithm	Over Shoot (%)	Settling Time (sec)	Rise Time (sec)
GWO (Proposed)	1.8	0.244	0.167
IWO [5]	7.16	1.95	0.493
PSO [5]	25.5	2.38	0.409
SFS [7]	0	1.06	0.638

Similarly, the comparison of step responses has been shown in Fig. 9 and comparative analysis has been shown in terms of response's parameters for operating point 3 in Table 7.

Table 7: Comparison of response parameters for operating point 3 ; $R_a = 0.1$; $K = 0.014$

Algorithm	Over Shoot (%)	Settling Time (sec)	Rise Time (sec)
GWO (Proposed)	2.18	0.384	0.145
IWO [5]	9.33	1.7	0.428
PSO [5]	27.2	2.06	0.365
SFS [7]	0.438	0.852	0.539

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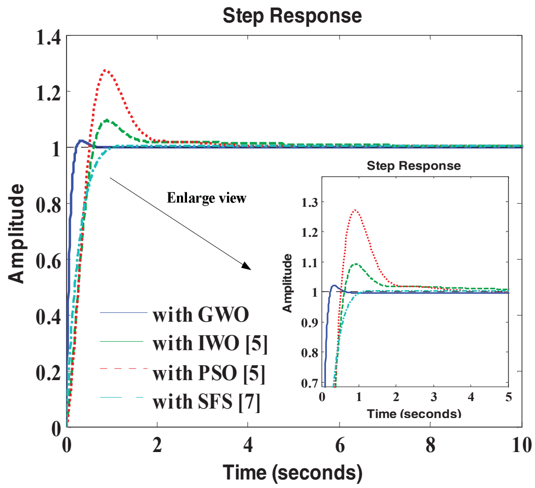

Fig. 9. Comparison of step responses for Operating point 3; $R_a = 0.1$; $K = 0.014$

Also, for operating point 4, the comparison of step responses has been shown in Fig. 10 along with the comparative analysis of response's parameters in Table 8.

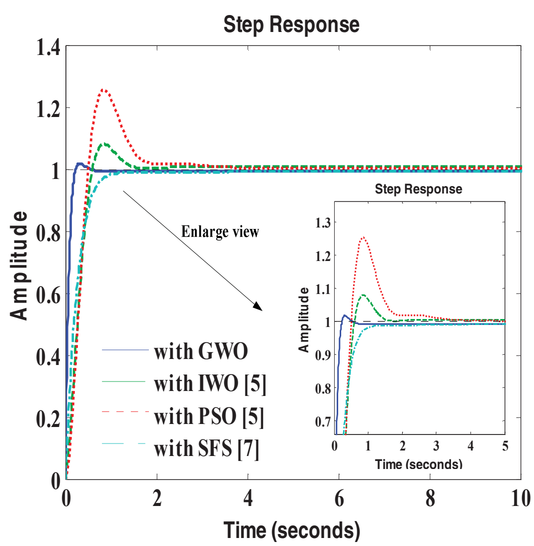

Fig. 10. Comparison of step responses for Operating point 4; $R_a = 0.3$; $K = 0.015$

The complete robustness analysis of the DC motor with PID controller tuned by GWO with ITAE as an objective function for all operating points has been shown in Figure 13. It can be seen in Fig. 11 that, step response of DC motor for all operating points is matching, i.e. there is no affect of variations in the parameters of DC motor on the performance of PID controller once it is tuned by GWO.

Table 8: Comparison of response's parameters for operating point 4; $R_a = 0.3$; $K = 0.015$

Algorithm	Over Shoot (%)	Settling Time (sec)	Rise Time (sec)
GWO (Proposed)	1.74	0.203	0.138
IWO [5]	7.92	1.32	0.414
PSO [5]	25.3	1.83	0.353
SFS [7]	0	0.968	0.53

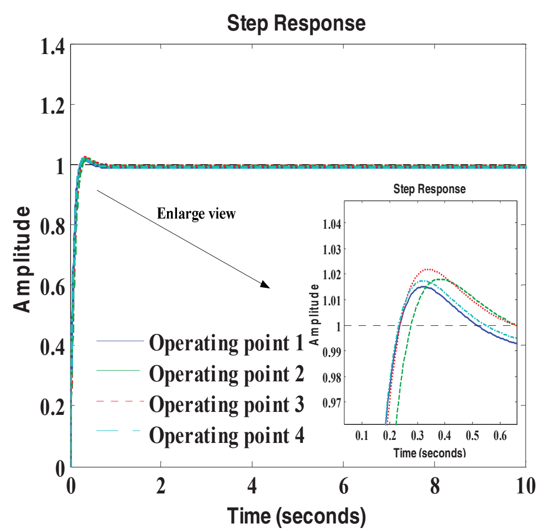

Fig. 11. Speed comparison of DC motor for all operating points

VII. CONCLUSIONS

The application of GWO algorithm in optimal speed control of DC motor has been shown. The ITAE has been taken as an objective/fitness function. Comparison of proposed GWO/PID scheme with ITAE has also been shown with other existing techniques; such as IWO/PID [5], PSO/PID [5] and SFS [7]. The simulation results reveal that GWO/PID scheme with ITAE as an objective function gives comparable overshoot and other parameters such as; settling time and rise time are less in comparison to existing approaches. The robustness analysis of proposed GWO/PID scheme has also been carried out with variations in the parameters of DC motor along with comparative analysis. It

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has been observed that, there is no affect of variations in the parameters of DC motor on the performance of PID controller.

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Appendix

The Pseudo codes of GWO algorithm has been given as follows [8] :

Initialize the algorithm parameters and generate the initial populations (positions of the wolves or agents)
Determine the fitness of each agent
Estimate X_α, X_β and X_δ, the position of α, β and δ wolves (the three best search agents)
while(t<Max number of iterations)
for each search agent
Update the position of current search agent
end for
Update search agents
Calculate the fitness of all search agents
Update the position of α, β and δ wolves
Increase the iteration count
end while
Display the best wolves X_α

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